EXPERIMENT NO. 1: LOCAL LIBRARY MANAGEMENT SYSTEM

AIM:

To design and develop a responsive web application for a Local Library Management System that enables book catalog searching, borrowing and return management, fine calculation, and administrative control over books and semester materials.

PSEUDOCODE (IN 6 SIMPLE POINTS):

Step 1: Initialize the system with catalog data containing book titles, authors, categories, and availability status.

Step 2: Display an intuitive user interface with navigation tabs for Book Catalog, Student Borrow Dashboard, Fine Calculator, and Admin Portal.

Step 3: Implement a real-time search filter allowing users to search books by title, author, or semester-wise study material.

Step 4: Record student borrowing transactions by capturing student ID, book title, issue date, and setting an automated return due date.

Step 5: Calculate overdue fines dynamically by determining days elapsed past the due date multiplied by the daily fine rate.

Step 6: Enable librarian administrative functions to add new books, update/delete existing records, and upload semester study materials.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Local Library Management System</title>
  <style>
    body { font-family: 'Segoe UI', Arial, sans-serif; margin: 0; background: #f0f2f5; color: #333; }
    header { background: #1a365d; color: white; padding: 18px 30px; display: flex; justify-content: space-between; align-items: center; }
    .container { max-width: 1000px; margin: 25px auto; padding: 20px; background: white; border-radius: 8px; box-shadow: 0 2px 10px rgba(0,0,0,0.1); }
    .tabs { display: flex; gap: 10px; margin-bottom: 20px; border-bottom: 2px solid #e2e8f0; }
    .tab-btn { padding: 10px 20px; cursor: pointer; border: none; background: none; font-weight: 600; color: #4a5568; }
    .tab-btn.active { color: #2b6cb0; border-bottom: 3px solid #2b6cb0; }
    .panel { display: none; }
    .panel.active { display: block; }
    input, select, button { padding: 10px; margin: 6px 0; border: 1px solid #cbd5e0; border-radius: 4px; box-sizing: border-box; }
    button { background: #2b6cb0; color: white; border: none; cursor: pointer; font-weight: bold; }
    button:hover { background: #1a4971; }
    table { width: 100%; border-collapse: collapse; margin-top: 15px; }
    th, td { border: 1px solid #e2e8f0; padding: 10px; text-align: left; }
    th { background: #edf2f7; color: #2d3748; }
    .fine-badge { background: #fed7d7; color: #9b2c2c; padding: 4px 8px; border-radius: 4px; font-weight: bold; }
  </style>
</head>
<body>
  <header>
    <h2>Central Library Portal</h2>
    <span>Role: Student / Admin</span>
  </header>
  <div class="container">
```

```
    <div class="tabs">
      <button class="tab-btn active" onclick="showTab('catalog')">Book Catalog</button>
      <button class="tab-btn" onclick="showTab('borrow')">Borrow & Return</button>
      <button class="tab-btn" onclick="showTab('fine')">Calculate Fine</button>
      <button class="tab-btn" onclick="showTab('admin')">Librarian Admin</button>
    </div>

    <!-- Catalog Tab -->
    <div id="catalog" class="panel active">
      <input type="text" id="searchBox" placeholder="Search by Book Title or Subject..." onkeyup="filterBooks()"
style="width: 100%;">
      <table id="bookTable">
        <thead><tr><th>ID</th><th>Book Title</th><th>Author</th><th>Semester</th><th>Status</th></tr></thead>
        <tbody>
          <tr><td>B101</td><td>Data Structures & Algorithms</td><td>Mark Allen</td><td>Sem
3</td><td>Available</td></tr>
          <tr><td>B102</td><td>Internet Programming Essentials</td><td>Paul Deitel</td><td>Sem
5</td><td>Available</td></tr>
          <tr><td>B103</td><td>Database System Concepts</td><td>Silberschatz</td><td>Sem
4</td><td>Issued</td></tr>
        </tbody>
      </table>
    </div>

    <!-- Borrow & Return Tab -->
    <div id="borrow" class="panel">
      <h3>Issue Book to Student</h3>
      <form id="issueForm" onsubmit="issueBook(event)">
        <input type="text" id="studentId" placeholder="Student Roll No" required style="width: 48%;">
        <input type="text" id="issueBookId" placeholder="Book ID (e.g., B101)" required style="width: 48%;"><br>
        <label>Issue Date: </label><input type="date" id="issueDate" required>
        <label>Return Due Date: </label><input type="date" id="dueDate" required><br>
        <button type="submit">Issue Book</button>
      </form>
      <div id="issueStatus" style="margin-top: 15px; color: green; font-weight: bold;"></div>
    </div>

    <!-- Fine Calculator Tab -->
    <div id="fine" class="panel">
      <h3>Overdue Fine Calculation</h3>
      <p>Fine Policy: Rs. 5 per day after due date.</p>
      <label>Due Date: </label><input type="date" id="calcDueDate">
      <label>Actual Return Date: </label><input type="date" id="calcReturnDate"><br>
      <button onclick="calculateFine()">Check Fine</button>
      <div id="fineResult" style="margin-top: 15px;"></div>
    </div>

    <!-- Admin Tab -->
    <div id="admin" class="panel">
      <h3>Add New Book to Inventory</h3>
      <input type="text" id="newTitle" placeholder="Book Title" style="width: 48%;">
      <input type="text" id="newAuthor" placeholder="Author Name" style="width: 48%;"><br>
      <select id="newSem" style="width: 48%;">
        <option value="Sem 1">Semester 1</option>
        <option value="Sem 3">Semester 3</option>
        <option value="Sem 5">Semester 5</option>
      </select>
      <button onclick="addBook()">Add Book to Catalog</button>
    </div>
  </div>

  <script>
    function showTab(tabId) {
      document.querySelectorAll('.panel').forEach(p => p.classList.remove('active'));
      document.querySelectorAll('.tab-btn').forEach(b => b.classList.remove('active'));
      document.getElementById(tabId).classList.add('active');
```

```
            event.target.classList.add('active');
        }

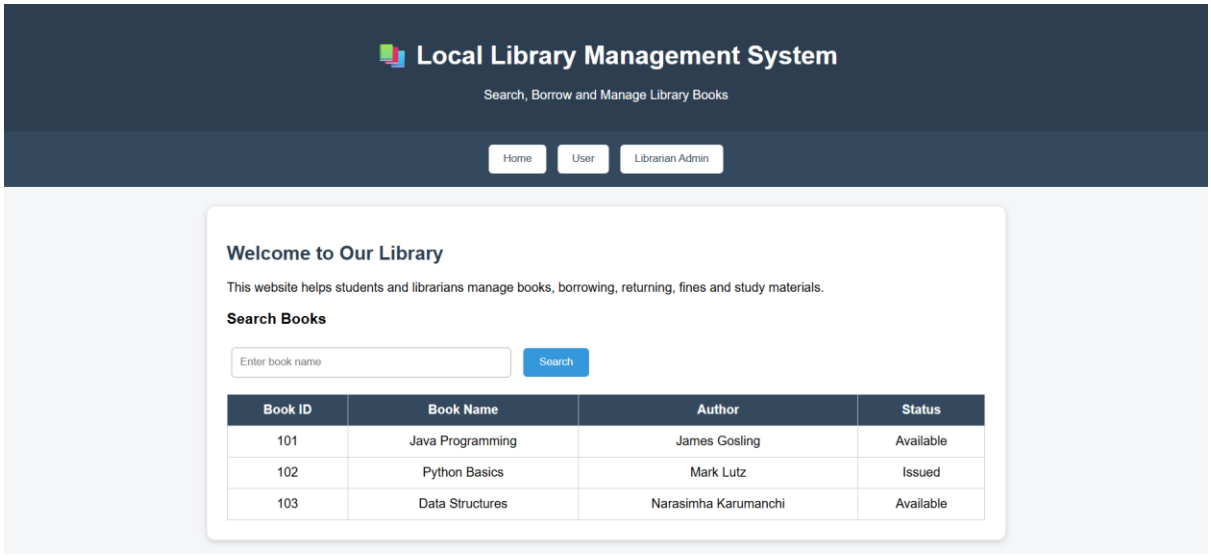
        function filterBooks() {
            let filter = document.getElementById('searchBox').value.toUpperCase();
            let tr = document.getElementById('bookTable').getElementsByTagName('tr');
            for (let i = 1; i < tr.length; i++) {
                let text = tr[i].innerText.toUpperCase();
                tr[i].style.display = text.includes(filter) ? '' : 'none';
            }
        }

        function issueBook(e) {
            e.preventDefault();
            let sid = document.getElementById('studentId').value;
            let bid = document.getElementById('issueBookId').value;
            let exp = document.getElementById('dueDate').value;
            document.getElementById('issueStatus').innerText = `Success: Book ${bid} issued to Student ${sid}. Return Due Date: ${exp}`;
        }

        function calculateFine() {
            let due = new Date(document.getElementById('calcDueDate').value);
            let ret = new Date(document.getElementById('calcReturnDate').value);
            let diffDays = Math.ceil((ret - due) / (1000 * 60 * 60 * 24));
            if (diffDays > 0) {
                let fine = diffDays * 5;
                document.getElementById('fineResult').innerHTML = `<span class="fine-badge">Late by ${diffDays} days. Fine Amount: Rs. ${fine}</span>`;
            } else {
                document.getElementById('fineResult').innerHTML = '<span style="color: green; font-weight: bold;">Returned on time! No fine applicable.</span>';
            }
        }

        function addBook() {
            let title = document.getElementById('newTitle').value;
            let author = document.getElementById('newAuthor').value;
            let sem = document.getElementById('newSem').value;
            if(!title || !author) return alert('Enter book details');
            let tbody = document.getElementById('bookTable').getElementsByTagName('tbody')[0];
            let row = tbody.insertRow();
            row.innerHTML = `<td>B${Math.floor(100+Math.random()*900)}</td><td>${title}</td><td>${author}</td><td>${sem}</td><td>Available</td>`;
            alert('Book added to inventory!');
            document.getElementById('newTitle').value = '';
            document.getElementById('newAuthor').value = '';
        }
    </script>
</body>
</html>
```

Output:



RESULT:

Thus, the web application for the Local Library Management System was successfully designed, developed, and verified with catalog search, book issue/return tracking, fine calculation, and administrative inventory controls.